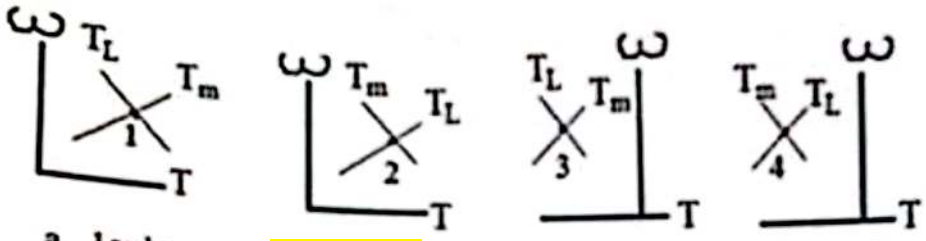
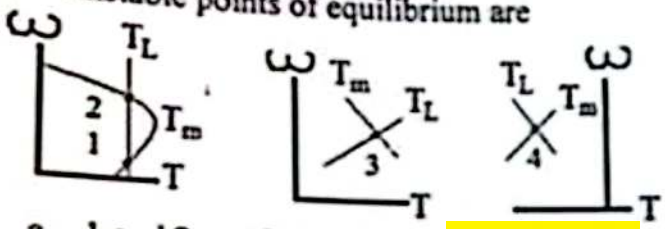
Answer the following questions by circling the right answer in the table at the end of this question.

1. Stable points of equilibrium is



a. 1only b. 2 only c. 3 only d. 2 and 4

2. Unstable points of equilibrium are



a. 1 and 2 b. 1 and 3 c. 1 and 4 d. 2 and 3

3. If the torque developed by the motor is $T = T_L + J\frac{d\omega}{dt}$, the drive is

a. decelerated b. accelerated c. working at S.S.P d. none of these

4. In electric drive system, the main function of the controller is

a. matching the motor and power converter to meet the load requirements
b. adapting to the changes of the environment only
c. sensing the feedback variables only
d. controlling the power converter only

5. Loads which require coupling with the motor shaft through transmission mechanical system (TMS) are

a. loads which need low speeds and high torques
b. loads which need high speeds and low torques
c. loads which need high speeds and high torques
d. loads which need low speeds and low torques

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6. The electric drive system with following motor torque and load torque expressions is stable:

a) $T = 10 + 2\omega, T_L = 10$

b) $T = 25 - 3\omega, T_L = 10$

c) $T = 15 + 4\omega, T_L = 15 + \omega$

d) $T = 25 + 5\omega, T_L = 2 + 2\omega^2$

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7. The drive system output power is:
a. Thermal power b. Electrical power
c. Mechanical power d. Electromagnetic power

8. Component determines the motor starting time is
a. motor inertia b. motor load c. speed change rate d. all of these

9. The DC motor that should never be started on no-load is
a. series b. compound
c. shunt d. permanent magnet

10. The active load torque industrial machinery is:
a. pump b. crane c. ventilator d. turning machine

11. Production machineries torque-speed characteristic is:
a. Decreasing linear characteristic b. Increasing non-linear characteristic
c. Decreasing non-linear characteristic d. Independent on speed

12. For D.C. series motor which of the following is incorrect
a. suitable for heavy duty starting b. torque varies as armature current
c. has non-linear characteristics d. has hard characteristics

13. Crane motor lowers load of 10 N.m through gearbox, which has gear ratio of 5. If the system efficiency equals 0.9, the referred load torque to motor shaft is:

a. 2.22N.m b. 50N.m c. 2N.m d. 1.8N.m

14. Electric motor has moment of inertia of 1kg.m2, connected through gearbox of 5 gear ratio to drum with moment of inertia 4 kg.m2. The drive system total (equivalent) moment of inertia equals:
a. 21 $kg.m^2$ b. 6$kg.m^2$ c. 1.16 $kg.m^2$ d. none of these

15. When varying load torque, separately excited dc motor has constant:
a. speed b. flux c. back emf d. armature current

16. If the motor works in the 4th quadrant. This mode of operation is a
a. reverse braking mode b. forward regenerative mode
c. motoring mode d. reverse motoring mode

17. Separately excited dc motor with the following ratings:
$P_{2N}=10kW, V_a=V_f=220V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A, k\Phi_N=2Vs$ The nominal motor torque is:
a. 50 N.m b. 100N.m c. 12.5N.m d. 27.5 N.m

18. Separately excited dc motor with the following ratings:
$P_{2N}=10kW, V_a=V_f=220V, \omega_N=100rad/s, R_a=0.4\Omega, I_{an}=50A, k\Phi_N=2Vs$ Nominal angular speed drop is:
a. 100 rad/s b. 20 rad/s c. 10 rad/s 80 rad/s

19. Plugging mode in the 4th quadrant is very dangerous mode due to overheating, but it used under nominal load in:
a. pump machinery b. drilling machine c. escalator d. crane

20. Adding resistance in series with armature of dc motor will cause:
a. Increasing of motor torque b. additional heat in the armature circuit
c. Increasing of short- circuit current d. Increasing of starting torque

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4. Absolutely hard speed – torque characteristics can be obtained in dc motor drive in closed – loop systems ( )
5. Squirrel – cage motors are the most widely used in industry ( )

Q3 : (5 marks)

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Calculate the starting time of the drive system with the following data:
J_{eq}=1okg.m2, T_m=15+0.5ω , T_L=5+0.6ω,
(consider final speed ω_2=0.95 ωss)
Solution

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